

Sabrina Adler

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Education

The Pennsylvania State University, State College, PA

May 2026

Integrated M.S. in Computer Science and Engineering

B.S. in Computer Science

B.S. in Mathematics (Systems Analysis)

Schreyer Honors College

GPA: 3.96

Research Experience

Master's Thesis Researcher

Jan 2025 – Present

Pennsylvania State University

CSE Advisor: Prof. Mehrdad Mahdavi • Ecology Advisor: Prof. Christina Grozinger

- Developing a scalable computer vision pipeline for pollinator habitat assessment from unstructured ground-level imagery, including segmentation, representation learning, and feature extraction.
- Formulating habitat quality estimation as a contrastive preference-learning problem, integrating ecological field knowledge with learned visual representations to produce continuous, interpretable quality scores.
- Designing quantitative proxies for ecological concepts (such as floral diversity, plant genera composition) and evaluating their statistical and biological validity.
- Research supported by NSF INSECT NET (Grant #2243979), an interdisciplinary graduate training program in pollinator biodiversity.

Research and Development Engineer Intern

May 2023 – Dec 2024

Applied Research Laboratory, Pennsylvania State University

State College, PA

- Designed and deployed a modular full-stack LLM serving platform (FastAPI, React) supporting multiple open-source foundation models.
- Developed and optimized Retrieval-Augmented Generation, achieving a 16× increase in effective context capacity and 105%+ improvement in response relevance through systematic evaluation of chunking, embeddings, and prompting strategies.
- Conducted in-depth evaluation of Transformer architectures and automated dataset generation pipelines to improve GPU efficiency, fine-tuning workflows, and model performance analysis.

Undergraduate Researcher, Insect Behavioral Ecology

Jan 2023 – May 2023

Pennsylvania State University

State College, PA

- Investigated aphid behavioral responses to predator exposure by comparing alarm reaction times following 24-hour exposure to lady beetle chemical cues versus no exposure.
- Designed controlled experiments to test hypotheses regarding habituation to predator-associated olfactory signals.
- Collected and analyzed behavioral data to evaluate changes in response latency and defensive signaling.

Teaching Experience

Instructional Contributor, CSE/ENT/ME 597: Technology and AI in Living Systems

Fall 2025

Pennsylvania State University

State College, PA

- Designed and delivered an assignment guiding students through end-to-end object detection: dataset collection and annotation (Roboflow), YOLOv8 fine-tuning in Jupyter, model evaluation, and deployment via a FastAPI inference endpoint integrated with a robotic system.
- Taught practical machine learning system design, including virtual environments, dependency management, modular model encapsulation, and API-based model serving.

Presentations

PolliSense: A Vision-Based Assessment of Pollinator Habitat Quality

Adler, S. (2026). International Conference on Pollinator Biology, Health, and Policy, Montana State University.

Constructing the Real Numbers via Dedekind Cuts

Adler, S. (2025). Penn State Math Club Undergraduate Lecture Series.

Neural Networks and Backpropagation: A Chalk Talk

Adler, S. (2024). Penn State Math Club Undergraduate Lecture Series.

The Christofides Algorithm and the Traveling Salesman Problem

Adler, S. (2023). Penn State Math Club Undergraduate Lecture Series.

Leadership and Academic Service

Penn State Math Club

President

2025 – 2026

Vice President

2023 – 2025

- Organized biweekly lectures featuring faculty and invited speakers, strengthening undergraduate engagement with advanced mathematics.
- Secured and developed a dedicated Math Club lounge as an academic community space, coordinating logistics, furnishing, and decor
- Established a shared mathematics textbook library and organized large-scale academic initiatives (Integration Bees, competition preparation groups, student–faculty dinners).
- Expanded institutional visibility through strategic digital outreach (8M+ cumulative views) in collaboration with the Eberly College of Science communications team.

Mathematics Directed Reading Program (DRP)

Founder and Undergraduate Head

2025 – 2026

- Established Penn State’s first Math Directed Reading Program pairing undergraduates with graduate mentors for semester-long guided readings.
- Designed application, matching, and evaluation process.
- Organized kickoff event and end-of-semester research symposium.
- Structured reading groups across advanced topic areas including topology, dynamical systems, Fourier analysis, differential geometry, game theory, and mathematical biology.
- Scaled the program to approximately 35 undergraduate mentees and 15 graduate mentors.

Industry Experience

Data Automation and Machine Learning Intern

Jun 2025 – Present

Merck & Co.

Upper Gwynedd, PA

- Developed a temporal CNN anomaly detection system for 300+ regulated controlled temperature units (CTUs), predicting temperature excursions 24 hours in advance (90%+ accuracy, 85%+ recall) within a pharmaceutical compliance framework.
- Designed and deployed scalable AWS-based machine learning infrastructure (SageMaker, S3, Lambda, EC2) with automated retraining pipelines, balancing statistical performance with regulatory and operational constraints.
- Translated predictive model outputs into actionable maintenance workflows by developing RESTful inference APIs, automated reporting systems, and alert notifications used by engineering and compliance teams.

Honors and Awards

- Euler Memorial Scholarship (2025) — Competitive merit award granted annually to two undergraduate mathematics majors.
- Russell D. and Alma D. Hartz Scholarship (2022–2026) — Merit scholarship awarded to undergraduates in the College of Engineering.
- Schreyer Honors College Academic Excellence Scholarship (2022–2026).
- Girl Scout Gold Award (2022).